

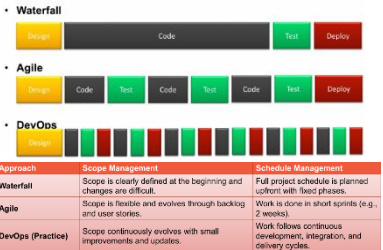
Project: a temporary effort/work undertaken to create a unique product, service, or result. **Difficulty:** uncertainty, hidden complexity. **Failed Reason:** requirement change, underestimated complexity, misalignment between business and technical teams. Core PM skill (Planning management)+Soft skill (communication).

Success means stakeholders satisfaction, align with strategic goals, team performance.

Triple Constraints: Scope ↑ → Time/Cost ↓ ; Time ↓ → Cost ↑ /Scope ↓ ; Budget ↓ → Scope ↓

Project Management (delivering a specific output) → **Program** Management (collection of related projects, deliver business value which could not get by managing them individually) → **Portfolio** Management (collection of programs and projects, alignment with organizational strategy)

Project Lifecycle (PMBOK 5 Process) is the structured sequence of phases a project goes through from start to finish. Execution has the biggest influence. Ensure consistency and predictability+Improve team collaboration+Manage risks, time, and costs+Align work with business goals. **Initiation** defines the project and evaluates feasibility for formal approval, include stakeholder identification, Expert judgement, SWOT. **Planning** builds a comprehensive guide plan, outputting the approved Project Management Plan. **Execution** performs the actual work to develop and review project deliverables. **Monitoring & Controlling** tracks performance to issue status reports and required course corrections. **Closure** wraps up the project to deliver results, yielding the final report, lessons learned, and official client sign-off.



Waterfall: linear, each phase must be completed before the next begins, not change, many documentation, not contact with customer. Simple+Clear milestone VS Not flexible+Not fit dynamic

Agile: not a management technique but a mindset, release by features. Only include Dev parts

4 Key Agile Values



DevOps: It is a culture. A set of practices that integrates software development and IT operations to *shorten the development lifecycle*, increase deployment frequency, and deliver high-quality software continuously. **Build** means integrate the code. **Automation** enforces behavior at scale. **CI:** Auto build and test tools such as



Jenkins. **Continuous Delivery:** ensuring the software is always in a releasable state. **Continuous Deployment:** your production pipeline is released to your customers' environment automatically.

Scrum VS Kanban (Both belong to Agile)

Scrum: Sprint/Product Backlog/Sprint Backlog/Increment

Kanban: continuous delivery, limiting work in progress (**WIP**), and workflow transparency.

Feature	Scrum	Kanban
Structure	Time boxed sprints	Continuous flow
Roles	Defined (PO, SM, Team)	No specific roles required
Workload	Planned per sprint	Pulled as capacity allows
Change Handling	Limited during Sprint	Change is allowed anytime
Visual Management	Optional board	Mandatory Kanban board
Best For	Teams needing structure	Teams needing flexibility

Requirement Gathering: Ensures the final product meets stakeholder needs; Reduces scope creep and rework; Forms the foundation for design, development, and testing; Facilitates better project planning and estimation. **Methods:** Interviews, Surveys,

Methodology	When Defined	Documentation Level	Change Flexibility
Waterfall	Beginning	Heavy	Low
Agile	Iteratively	Moderate	High
Kanban	Continuous	Light	Very High
DevOps	Continuous + Feedback	Light to Moderate	Very High

Workshops, Observation, Document Analysis.

Identify Stakeholder: All identified in early stage, but style and frequency is different. **Stakeholder** is anyone who impacts or is impacted by the project: positively or negatively. **Key activities:** Identify, Analyze, Prioritize, Engage, Monitor/Engage.

Aspect	Waterfall	Agile / Scrum	Kanban	DevOps
When Identified	Early, during initiation	Early + continuous refinement	Continuous	Continuous
Level of Formality	High (documents, approvals)	Moderate	Low to Moderate	Moderate
Stakeholder Involvement	Limited after requirements phase	Frequent (Sprint Reviews)	Ongoing feedback	Continuous + operational feedback
Communication Style	Structured, scheduled	Collaborative, iterative	Flexible, on-demand	Cross-functional, integrated
Change Handling	Formal change requests	Adapted each sprint	Anytime	Continuous improvement

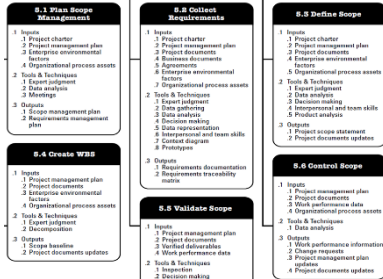
Project Charter: a formal document that authorizes the project and grants the project manager the authority to use organizational resources. **Includes:** Project title+end date+brief background.

Purpose: Establishes a clear vision of the project+Aligns stakeholders and secures executive support+Provides a high-level roadmap for planning+Act as a reference point for project.

Interest/Power Grid: **HP-HI (Manage Closely)** Project Sponsors who possess major authority and high investment, requiring continuous, intensive management; **HP-LI (Keep Satisfied)** individuals, such as Legal Compliance Officers, hold significant power but low daily interest, meaning they must be kept satisfied with high-level updates to prevent roadblocks; **LP-HI (Keep Informed)**, including End-users, lack direct authority but care deeply about outcomes, making regular information sharing essential to leverage them to your project advocates; **LP-LI (Monitor)** like General Suppliers, require minimal effort and should simply be tracked through low-frequency, routine communication.

1 Scope Management: defines how the project scope will be defined, validated, and controlled. A **WBS** item is the responsibility of only one individual, even though many people may be working on it. **Method: Analogous** (Review WBS of similar projects and tailor to your project), **Top-Down, Bottom-Up, Mind-mapping**

Issues: Scope Creeps (Uncontrolled changes or continuous growth in a project's scope without adjustments to time, cost, and resources), Miscommunication, Overambitious Scope.



2 Schedule Management: the process of planning, developing, managing, executing, and controlling a project's schedule so that it can be completed within the agreed time frame. **Processes:** 1 Plan Schedule Management, 2 Define Activities, 3 Sequence A, 4 Estimate A duration, 5 Developing S, 6 Control S

2. means break down work packages from the Work Breakdown Structure (WBS) into specific, manageable tasks.

3.1 Types of dependencies: **Mandatory** (cannot be changed, Hard logic), **Discretionary** (based on best practices), **External** (rely on

activities outside the project team's control. Example, Governance approval), **Internal** (within the project team's control); **3.2 AON** Activity on nodes, arrow show dependencies; **AOA** Activity on arrows, nodes show events.

3.3 Precedence Diagramming Method PDM: a Type of AON; **Finish to Start (FS)** - Activity B starts after A finishes (most common).

Start to Start (SS) - Activity B starts when A starts, namely, B cannot start until A start.

Finish to Finish (FF) - Activity B finishes when A finishes, namely, B cannot finish until A finish.

Start to Finish (SF) - Activity B finishes when A starts (rare). B cannot finish until A start.

3.4 Lead time: start next task earlier, Overlap between tasks. It accelerates the schedule.

3.5 Lag time: delay next task. It inserts a waiting period.

4.1 Three Point Estimate - PERT

- **Optimistic (O)** - The shortest time in which the task can be completed if everything goes well.
- **Most Likely (M)** - The best guess of the time required, assuming normal conditions.
- **Pessimistic (P)** - The longest time the task might take if things go wrong.

$$\text{Formula is: PERT} = \frac{(O+4M+P)}{6}$$

4.2 Analogous Estimating: top-down, based on historical data, fast, less cost, but require experts

4.3 Parametric Estimating: Require relationship between variables are consistent and scalable.

5.1 Gantt Chart: A horizontal bar chart; Visualizing the project timeline; Tracking progress against the baseline schedule; Identifying overlaps, leads, and lags; **length** represents its duration.

5.2 Critical Path Method: identify the longest sequence of dependent activities in the project. Any delay on the Critical Path will DELAY the entire project.

6.1 Fast Tracking Perform activities in parallel that were originally sequential. Increases risk but shortens schedule.

6.2 Crashing Add more resources to critical path activities to finish sooner. Increases cost, may increase risk.

3 Cost Management: process of estimating, budgeting, and controlling costs. 1 Plan C.M., 2 Manage Quality, 3 Control Quality

Why Critical? Rapidly changing technology, volatile pricing, Hidden costs (integration, security, compliance), Vendor lock-in & license renewals. **Challenges:** Hidden and recurring costs, Measurement challenges.

Type of Cost Tangible: Costs that an organization can easily measure in dollars (e.g., hardware purchases, software licenses, and team salaries). **Intangible:** Costs that are difficult to measure in monetary terms (e.g., impact on employee morale, loss of customer goodwill, or brand reputation). **Direct:** Costs that are directly related to producing the products and services of the project (e.g., project team travel expenses, project-specific materials). **Indirect:** Costs that are not directly related to the specific products or services but are indirectly related to performing the project (e.g., office rent, utilities, and general administrative support). **Sunk:** Money that has been spent in the past when deciding what projects to invest in. According to PMP standards, sunk costs are not included in the decision-making process for future project investments or current performance measurement.

Contingency Reserves: Funds for "known-unknowns" (partially planned situations). These are included in the cost baseline. Usually take 10% of total cost.

Management Reserves: Funds for "unknown-unknowns" (unpredictable situations). These are not included in the cost baseline.

1 Purpose: Ensures consistency, Provides guidelines for resource allocation and budget approvals, Helps prevent budget overruns.

2 Analogous, Parametric, Bottom-Up (most detailed), Three-Point, Expert Judgment

Benchmark: Using historical project data or industry averages to guide cost estimation.

Budget = Cost Estimates + Contingency Reserves + Management Reserves

Cost Baseline: time-phased budget; includes direct costs, indirect costs, and contingency reserves. No management reserves

4. Earned Value Management EVM: a methodology that integrates scope, schedule, and cost to objectively measure project performance and progress.

SV is Positive → ahead of schedule; Negative → behind schedule

CPI > 1 → cost efficient; CPI < 1 → cost overrun

SPI > 1 → ahead of schedule, efficient; SPI < 1 → behind schedule

Challenges of Applying EVM in Agile Projects: Changing scope makes Planned Value unstable; Story points and features are not directly measurable in dollars; Sponsors often expect cost forecasts in monetary terms rather than velocity metrics. **Solution:** Track

Term	Short Definition	Formula
Planned Value (PV)	Value of the work planned to be completed by a certain time	—
Earned Value (EV)	Value of the work actually completed	EV = PV of completed work
Actual Cost (AC)	Actual money spent to complete the work	—
Cost Variance (CV)	Difference between earned value and actual cost (budget performance)	CV = EV - AC
Schedule Variance (SV)	Difference between earned value and planned value (schedule performance)	SV = EV - PV
Cost Performance Index (CPI)	Measures cost efficiency of the project	CPI = EV / AC
Schedule Performance Index (SPI)	Measures schedule efficiency of the project	SPI = EV / PV
Estimate at Completion (EAC)	Estimated total project cost when the project finishes	EAC = BAC / CPI
Estimate to Complete (ETC)	Estimated cost required to finish remaining work	ETC = EAC - AC

Earned Business Value (EBV) instead of only cost efficiency, Map story points to cost using sprint burn rates, Recalculate EAC at the end of each iteration using rolling forecasts.

Traditional VS Agile Cost Management

Aspect	Traditional (Waterfall/Plan-driven)	Agile (Iterative/Adaptive)
Budgeting	Fixed upfront, detailed cost baseline	Incremental, funded per sprint/increase
Estimation	Comprehensive planning, WBS-based	Rolling-wave, evolving estimates
Scope Stability	Scope fixed early, baseline stable	Scope flexible, backlog reprioritized
EVM Application	Works well: PV, EV, AC tracked reliably (management)	Difficult: scope & metrics change frequently
Measurement of Value	Monetary units (dollars, hours, cost)	Story points, features, business value
Stakeholder reporting	Financial variance reports (CV, SV, CPI, SPI)	Velocity charts, burn-down, business value
Risk handling	Reserves set upfront (contingency, management)	Flexibility built into scope & incremental funding

4 Quality Management: Identifying the quality requirements and standards for the project and deliverable. About both product quality and process quality. **ISO9001** quality, **ISO/IEC27001** Security

Major principles: **Customer focus:** End-user satisfaction is the ultimate quality measure. **Prevention over Inspection:** It's cheaper to prevent defects than fix them later. **Continuous Improvement:** Using lessons learned and feedback loops (Agile, Kaizen, DevOps).

Fact-based decision making: Using metrics, dashboards, and data analysis. 1 Plan Q.M., 2 Manage Q, 3 Control Q

Project Manager, need to Monitor CoQ (Cost of Quality), **CoC** (Cost of Conformance), and **CoNC** (Cost of Non-Conformance):

CoQ=CoC+CoNC

Aspect	Cost of Conformance	Cost of Non-Conformance
Definition	The cost of ensuring that a product or service meets quality requirements	The cost incurred when the product fails to meet quality standards
Focus	Prevent defects proactively	Reactively fix defects
Components	Prevention + Appraisal	Internal + External Failures
IT Example	Automated tests, code reviews, training	Bug fixes, downtime, customer complaints
Goal	Minimize future costs and improve reliability	Minimize impact of failures

Quality Assurance Team: Define processes, methods, and metrics. **Quality Control Team:** Execute tests

Development Team: Follow coding standards, guidelines, and defined processes.

Agile Based QMP: Focuses on delivering quality incrementally. **DevOps based QMP:** Focuses on ensuring continuous quality from end-to-end. Automation every where

2.1 Process Audits: check if teams are following defined quality standards, policies, and procedures. catch process issues before they lead to product defects.

2.2 Quality Assurance Reviews: the process of structured evaluations of deliverables and processes, performed throughout the project. Includes peer reviews, design walkthroughs, and retrospectives.

2.3 Continuous Process Improvements: an ongoing effort to refine processes for better efficiency, effectiveness, and adaptability. Often guided by PDCA (Plan-Do-Check-Act) or Six Sigma DMAIC.

3.1 Verification: "Did we build it right?" (meets specs).

3.2 Validation: "Did we build the right thing?" (meets user needs).

Unit testing: Tests individual components or modules of the software in isolation.

Integration testing: Tests the interaction between multiple modules or components.

System testing: Tests the complete integrated system against requirements.

User Acceptance testing: Final testing performed by end-users or clients.

Tools : **Cost-Benefit Analysis**, **Cause and Effect Diagram**, **Quality Control Chart**, **Checksheet**, **Flowcharts**, **Scatter Diagram**, **Histogram**, **Pareto** (80% of problems are caused by 20% of the causes), **Lean** (Focuses on eliminating waste: e.g., redundant processes, delays in defect resolution, excessive documentation), **Six Sigma** (Focuses on reducing variation and defects using data-driven approaches (DMAIC - Define, Measure, Analyze, Improve, Control)), **Deming's** 14 Points (philosophy, Focusing on process improvement, not just inspection, provides principles and guidelines, ensure the QMP is process-focused, preventive, and continuous.), **Juran's** 10 Steps (practical, Quality objectives are clear

and measurable, a structured framework for ensuring that the plan is actionable, systematic, and management-led.)

Trade off: Cost vs Quality: Higher quality often requires more investment (prevention, testing, training). Cutting costs can increase defects and rework. **Time vs Quality:** Tight schedules may force reduced testing or shortcuts, risking quality. Extending timelines allows thorough QA but may increase costs. **Scope vs Quality:** Adding more features can dilute focus on quality of existing functionality. **Prevention vs Quality:** Investing in prevention (training, standards) reduces defects later.

CMMI Capability Maturity Model Integration: • Level 1: Initial - unpredictable, ad hoc. • Level 2: Managed - processes are project-specific. • Level 3: Defined - processes are standardized across the org. • Level 4: Quantitatively Managed - metrics driven processes. • Level 5: Optimizing - continuous improvement.

5 Resource Management: the process of identifying, acquiring, and managing resources. 1. Plan R.M., 2. Estimate Activity Resou, 3

Acquire Res, 4 Develop Team, 5 Manage Team, 6 Control Res Leadership → Inspiring, motivating, enabling collaboration.

Autocratic (Directive): Leader makes decisions unilaterally. Pros: Fast decision-making, useful in crises (e.g., system outage). Cons: Low motivation, stifles creativity. **Democratic** (Participative): Involves the team in decision-making. Pros: Encourages innovation, increases engagement. Cons: Slower decisions, risk of conflict. **Laisses-faire** (Hands-off): Leader provides minimal direction, gives autonomy. Pros: Works with highly skilled, self-motivated teams. Cons: Can lead to confusion or lack of coordination.

Management → Planning, organizing, controlling resources.

Aspect	Intrinsic Motivation	Extrinsic Motivation
Definition	Motivation that comes from within the individual: driven by personal satisfaction, curiosity, or passion	Motivation that comes from external factors: rewards, recognition, or avoidance of punishment
Characteristics	• Enjoyment of the work itself. • Desire to learn or master a skill. • Feeling of accomplishment.	• Driven by tangible or social incentives. • Performance linked to external rewards.
Source of drive	Internal (self-driven)	External (reward/punishment)
Focus	Enjoyment, mastery, curiosity	Money, recognition, promotions

Motivation is the internal or external drive that influences people's behavior, effort, and persistence in achieving goals.

Maslow hierarchy of needs Theory



Human needs are arranged in a hierarchy, where lower-level needs must be met before higher-level needs motivate behavior.

In IT project management, understanding these needs helps leaders motivate and retain project team members.

Motivators (intrinsic, job context).

- Relate to the work itself.
- When present → increase job satisfaction and motivation.
- When absent → do not necessarily cause dissatisfaction, but lead to lack of motivation.

Hygiene Factors (extrinsic, job context).

- Relate to the work environment.
- When absent → cause dissatisfaction.
- When present → do not increase motivation, but prevent dissatisfaction.

RACI Matrix: a responsibility assignment tool that clarifies roles and responsibilities for project tasks or deliverables. **Responsible:** Person(s) who do the work or complete the task. **Accountable:** Person who is ultimately answerable; approves work and ensures task completion. **Consulted:** People whose input is sought before the task is done (two-way communication). **Informed:** People who need to be updated on progress or decisions (one-way communication).

Resource Breakdown Structure: Level 1: Broad categories of resources (Human, Equipment, Software, Materials) Level 2: Subcategories (for humans → roles; for software → tools/platforms. Level 3: Specific resources (names, tools, versions, servers, licenses)

Resource Histogram: a bar chart that shows the amount of time each resource is allocated over a period of time. X-axis: Time periods (days, weeks, months, sprints). Y-axis: Resource usage (hours, FTEs, or number of resources). Bars: Show how much of each resource is allocated in a given period.

2 Estimating Activity resources: Expert judgement, Analogous estimating, Bottom-up, Three-point estimating (PERT).

3.1 Acquiring Resources: Internal: Use existing employees or in-house resources. Familiar with company culture, faster onboarding. **External:** Hire new employees, contractors, or consultants. Brings specific skills, expertise, or additional capacity.

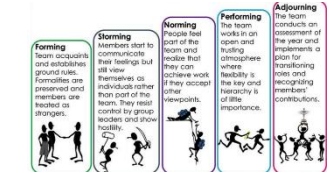
Procurement: Acquire equipment, software, or services through

vendors, **Multi-source** - Combination of internal resources, contractors, and vendors.

3.2 Resource Load = $\sum$ Allocated Hours/Available Hours

3.3 Resource Leveling: the process of resolving resource over-allocation by adjusting the project schedule or redistributing resources. Ensure that no resource is overburdened. Methods include delay tasks, split tasks, and reassign tasks.

4.1 Tuckman's Model:



Mode	Description	Assertiveness	Cooperativeness	When to Use
Competing	Assert own viewpoint at the expense of others (win-lose)	High	Low	Critical issues needing quick decisions (limited time)
Collaborating	Work with others to find a win-win solution	High	High	Complex issues requiring input from all parties
Compromising	Find middle ground (partial win-lose)	Moderate	Moderate	Time-constrained situations or temporary alliances
Avoiding	Ignore or postpone the conflict	Low	Low	Minor issues or when emotions are high
Accommodating	Yield to others' needs (lose-win)	Low	High	Preserve relationships or when issues matter more to others

6.1 Productivity: how efficiently resources (human, equipment, or materials) are used to achieve project objectives.

6.2 Resource availability: extent to which resources (human, equipment, or materials) are accessible and ready for project tasks when needed.

Stephen Covey's 7 Habits provide a personal and team effectiveness framework. For HR.

Be Proactive	Begin with the end in mind	Put first things first
Think win-win	Seek first to understand, then to be understood	Synergize

6 Risk Management: Risk is about uncertainty: something that may or may not happen. Risk matters because: IT project is dynamic due to rapid change. **1plan risk management, 2identify risks, 3qualitative risk analysis, 4quantitative, 5plan risk response, 6monitor and control risk**

Goal: find risk early, minimize the risk probability, enable informed decision making under uncertainty.

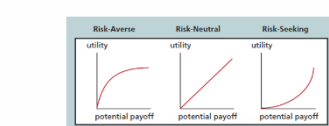
Benefits: increase the success rate, better decision making, cost and time saving.

Aspect	Positive Risks	Negative Risks
Context	These are opportunities	These are threats
Definition	Events that, if they occur, have a beneficial effect on project objectives.	Events that, if they occur, have a harmful effect on project objectives.
Impact	Beneficial for the project	Harmful for the project
Management Approach	Exploit, enhance, share, or accept.	Avoid, mitigate, transfer, or accept.

Risk Appetite: degree of uncertainty an organisation is willing to accept. **risk-seeking**: take bold moves to innovate. **risk-averse**: prioritise stability and predictability.

X-axis: Monetary value (or project benefit).

Y-axis: Utility (satisfaction or value derived).



Risk-Neutral Culture: balances opportunities and threats pragmatically. focus only on expected monetary value EMV **2. Identify Risk tools: Brainstorming, Delphi** obtaining consensus from a panel of experts through multiple rounds of anonymous surveys; Interviewing with stakeholders: **SWOT** (Strength, Weakness, Opportunities, Threats): assess internal strengths & weaknesses, and external opportunities & threats, identify both positive and negative risks.

3.1 A subjective method of prioritizing risks based on their probability of occurrence and impact on objectives.

3.2 Probability Impact Matrix: A grid mapping probability vs. impact. Upper right corner is the most important risk.

4.1 A numerical, data-driven process of analyzing the effect of risks on overall project objectives (cost, schedule, scope, quality).

4.2 Monte Carlo Simulation: A probabilistic technique that uses random sampling to simulate different possible outcomes of a project.

4.3 Decision Tree Analysis: Used to calculate **EMV = Possibility * Outcome** (Positive means Opportunity, Negative means Threat)

5.1 Risk responses for Negative Risks: Avoidance (changing project scope), **Mitigation** (Reduce probability or impact; extra testing), **Transfer** (buy insurance), **Acceptance** (Do nothing proactively), **Escalation** (report to leader)

5.2 Risk responses for Positive Risks: Exploitation: Ensure the opportunity happens. **Enhancement**: Increase probability or impact, ex. team training. **Sharing**: Partner with a third party to capture the benefit. Such as Collaborate with another firm for joint product launch. **Acceptance**: Take advantage if opportunity arises but don't actively pursue it. **Escalation**

5.3 Risk Register: A central **document** in risk management that records all identified risks, their characteristics, and planned responses. Updated throughout the project lifecycle. Ensures risks are visible, monitored, and assigned owners.

6. Traditional (Waterfall): Risks are identified, analyzed, and planned upfront during project initiation and planning. Risk register + formal reviews are created. **Agile**: Risks are addressed iteratively throughout the project in short cycles (sprints/iterations). Risk management is embedded in daily activities, not just upfront. Adv: Fast, adaptability to changing requirements. **Iterative Risk Reviews**: Risks are revisited at the end of every sprint (retrospectives). **Risk-Based Backlog Prioritization**:

User stories and features with higher uncertainty/risk are delivered earlier ("fail fast" principle). **Risk Burn-Down Charts**: Similar to sprint burn-down charts but track risk exposure over time. **Collaborative Ownership**: Risks are not managed by a single "risk manager" but shared among the team. **Hybrid**: Combines structured traditional risk processes (register, probability-impact matrix) with Agile practices (daily monitoring, adaptive responses). Adv: Balances flexibility with formal governance.

7 Procurement Management: the process of acquiring goods, services, or results from outside the project team to meet project needs. deciding what to buy, how to buy it, who to buy it from, and how to ensure the contract is successfully completed. **1Plan, 2Conduct** (obtaining seller responses, evaluating them, and awarding the contract), **3Control** (managing relationships with vendors, monitoring contract performance, and making changes as needed to ensure that both the buyer and seller meet contractual obligations.).

Statement of Work (SOW) is a detailed description of the products, services, or results to be delivered under a contract. It sets the scope, deliverables, timelines, and standards so that both the buyer and seller understand exactly what is expected.

Request for Proposal (RFP) is a formal document issued by a **buyer** (the project team or organization) to invite potential sellers/vendors to submit proposals on how they would meet the project's requirements. It provides enough detail about the project so vendors can propose solutions, costs, schedules, and approaches. Purpose: communicate requirements, encourage competition, evaluate proposals.

Request for Quotation (RFQ) is a formal procurement document issued by a **buyer** to solicit price quotations from suppliers for well-defined goods or services. primarily focused on price, delivery terms, and payment conditions when the requirements are clear

Aspect	Request for Quotation	Request for Proposal
Definition	A document to solicit price quotations from suppliers for well-defined goods or services.	A document issued by a buyer to invite potential sellers/vendors to submit proposals on how they would meet the project's requirements.
Focus	Price & delivery	Technical solution + price
Requirements	Well defined, standardized	Complex, needs innovation
Evaluation criteria	Cost driven	Quality, approach, and capability
When to use?	When you need price quotes for well-defined items.	When you want solutions and vendor creativity.
Typical use case	Commodities, shared items	IT projects, consulting, custom systems

and standardized. To obtain competitive pricing. To identify the most cost-effective supplier. To establish a baseline for vendor comparison. Feature: Standardized Products/Services, Price-Centric Evaluation, Detailed Specifications, Short Decision Cycle (compared to RFP).

Aspect	Fixed-Price contracts	Cost-Reimbursable contracts	Time and Material contracts
Definition	Price is agreed upon in advance, regardless of actual costs.	Buyer reimburses seller for actual costs, plus a fee/profit.	Hybrid of fixed and cost-reimbursable. Buyer pays per hour/day for materials.
Risk on Buyer	Low	High	Medium
Risk on Seller	High	Low	Medium
Advantages	Predictable cost for buyer, risk on seller.	Flexibility, good for R&D or uncertain projects.	Simple to administer, flexible scope.
Disadvantages	Seller may cut corners if cost rises, change requests are expensive.	Higher cost risk for buyer, no cost incentive for seller to control costs.	Risk of cost overruns if not monitored.
Best use case	Clear scope, predictable deliverables	R&D, uncertain requirements	Staff augmentation, flexible scope

3 types of contracts: Fixed price contracts: Firm Fixed Price (FFP): Price is final, no adjustments. **Fixed Price Incentive Fee (FPIF):** Price fixed, but with performance incentives. **Fixed Price with Economic Price Adjustment (FP-EPA):** Price fixed, but adjusted for inflation/market changes. **Cost reimbursable contracts: Cost Plus**

Fixed Fee (CPFF): Seller reimbursed + fixed fee. **Cost Plus Incentive Fee (CPPIF):** Seller reimbursed + incentive based on performance. **Cost Plus Award Fee (CPAF):** Seller reimbursed +

Buyer risk		Seller risk	
high	low	low	high
CPIC: Cost plus incentive fee	CPPIF: Cost plus incentive fee	CPAF: Cost plus award fee	FP-EPA: Fixed-price economic price adjustment
Onshore	Vendor in same country	Local software development company	
Nearshore	Vendor in nearby country/time zone	Indonesian vendor for Australian company	
Offshore	Vendor in distant country	India or Vietnam outsourcing for that company	
Hybrid	Mix of internal and external resources	Core strategy in-house, cloud offshoring	
Multisource	Mix of the different models of sourcing	A company develops its mobile app using its internal software development team instead of hiring an external vendor.	
Insourcing	Opposite of outsourcing. Movement of business process within the company but to another internal entity.		

award based on buyer's satisfaction. **Time & materials contracts. Vendor Evaluation Methods: Weighted Scoring Model** Best for Complex procurements where multiple factors matter, not just price. **Checklist**: Best for When compliance to minimum standards (e.g., certifications, licenses) is critical. **Vendor Scorecard**: Best for Long-term supplier relationships.

Outsourcing: Complete, Partial, Business Process (BPO): Outsourcing of non-core business processes related to the project. Example: Customer support, payroll processing, or IT helpdesk services. **IT/Technical**: Hiring specialized vendors for software development, cloud infrastructure, or cybersecurity.

8 Communication management: 1. Plan, 2 Manage, 3 Monitor. Leadership and governance also mention in 5.

- Leadership – People & Direction
 - Leaders set the vision, motivate teams, and manage change.
 - Without leadership, IT projects risk low morale, poor communication, and mismanaged expectations.
- Governance – Oversight & Accountability
 - Governance defines roles, responsibilities, structures, and decision-making mechanisms.
 - Without governance, projects face scope creep, poor accountability, and unmanaged risks.
- Alignment – Strategic Fit & Value Delivery
 - Alignment ensures IT projects are connected to business strategy and outcomes.
 - Without alignment, IT projects risk delivering tech that doesn't solve business problems.

Leaders provide vision and influence, while governance ensures checks and balances. Captain VS Navigate Sys VS Destination.

Aspect	Project Leadership	Project Governance
Definition	Project leadership is the application of knowledge, skills, tools, and techniques to plan, execute, and control projects so that they meet specific objectives within scope, time, cost, and quality constraints.	Project leadership is the ability to inspire, guide, and influence a project team and stakeholders to achieve the project's objectives, beyond just managing tasks and schedules.
Characteristics	- Task-oriented and process-driven. - Ensures compliance with governance frameworks and standards. - Focuses on efficiency, resource allocation, and deliverables.	- People-oriented and vision-driven. - Builds team cohesion and trust while providing direction and motivation during uncertainty. - Focuses on long-term value, adoption, and transformation.
Focus	Tasks, processes, tools	People, vision, influence
Goal	Deliver on scope, time, cost, quality	Inspire, align, and drive adoption & value
Approach	Control, planning, monitoring	Motivation, guidance, adaptability
Orientation	Short-term efficiency	Long-term impact

Aspect	Project Leadership	Project Governance
Focus	People, vision, and inspiration.	Processes, tasks, and control.
Orientation	Fulfill-oriented asks "Where are we going? Why does it matter?"	Process-oriented asks "How do we get it done efficiently and correctly?"
Approach	Motivates, influences, and empowers the team.	Plans, organizes, monitors, and controls the team.
Style	Flexible, adaptive, and willing to take risks.	Structured, methodical, and risk-averse.
Outcome	Creates engagement, innovation, and long-term value.	Delivers outputs on time, within budget, and per scope.

Governance Structure: Steering Committees (group of senior stakeholders, ensure aligned with organizational goals), Project Management Offices (**PMO**, centralized organizational unit or team bridge between project teams, leadership, and stakeholders), **Project Sponsors** (a senior executive or stakeholder, provides resources, link between the project team and the organization's strategic leadership).

Governance mechanisms: Steering Committee, PMO, Sponsor, Phase-Gate reviews, Performance reporting and dashboards.

Governance challenges: Ambiguity in decision-making. Project information is not timely, accurate, or clear, Projects are executed without clear alignment to organizational goals.

Strategic Alignment: Ensuring the project supports the organization's long-term goals, mission, and vision. **Business-IT Alignment**: Ensuring IT solutions deliver real business value and not just technical outputs. **Stakeholder Alignment**: Ensuring that the expectations of executives, end-users, customers, and regulators are properly managed and satisfied. **Team Alignment**: Ensuring project teams share a common understanding of objectives, roles, and priorities.

Why alignment matters in IT Projects? Maximizes Business Value, Reduces Wasted Effort, Improves Stakeholder Satisfaction, Boosts Adoption

Frameworks for Alignment: SAM (Strategic Alignment Model; Strategic Fit + Functional Integration; Business strategy, IT strategy, Organisational infrastructure and processes, IT

infrastructure and processes); **BSC** (Balanced Scorecard, integrates financial + non-financial measures); **COBIT** (Control Objectives for Information and Related Technology, IT governance and management framework by ISACA); **TOGAF** (The Open Group Architecture Framework; An enterprise architecture framework that aligns IT architecture with business strategy.)

value stream is the sequence of activities that an organization undertakes to deliver value (a product or service) to a customer. Focuses on end-to-end value delivery, rather than siloed departmental outputs. **Operational Value Streams** – How the business delivers value to end customers (e.g., customer support, product delivery). **Development Value Streams** – How the organization develops the systems and software that enable operational value streams

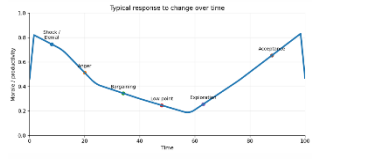
9 Stakeholder Management: Identify, engage, and manage stakeholders to meet expectations. **1. Identify stakeholders, 2 plan engagement, 3 manage engagement, 4 monitor engagement, 10 Integration Management**: coordination of all processes and activities across the project lifecycle to ensure the project meets its objectives. **1 Develop Project Charter, 2 Develop Project Management Plan, 3 Direct and Manage Project Work, 4 Manage Project Knowledge, 5 Monitor and Control Project Work, 6 Perform Integrated Change Control, 7 Close Project or Phase**

	Initiating	Planning	Executing	Monitoring & Controlling	Closing
Scope Management	✓	✓	✓	✓	✓
Schedule Management	✓	✓	✓	✓	✓
Cost Management	✓	✓	✓	✓	✓
Quality Management	✓	✓	✓	✓	✓
Resource Management	✓	✓	✓	✓	✓
Communication Management	✓	✓	✓	✓	✓
Risk Management	✓	✓	✓	✓	✓
Procurement Management	✓	✓	✓	✓	✓
Stakeholder Management	✓	✓	✓	✓	✓
Integration Management	✓	✓	✓	✓	✓

Tools: Project Management Information Systems (PMIS): e.g., MS Project, Jira, Asana, or Trello. **Dashboards**: Real-time updates on progress, risks, costs, and resources. **Change Control Boards (CCB)**: Teams that approve/reject project changes. **Documentation & Knowledge Repositories**: Wikis, Confluence, or SharePoint for lessons learned and best practices

Change management: Not belong to PMBOK

The Kübler-Ross Change Curve



Aspect	Project Management	Change Management
Focus	Focuses on scope, schedule, budget and quality	Focuses on communication, engagement and resilience
Orientation	Builds and launches the solution	Builds stakeholder confidence, training and support
Primary question: "Did we deliver it well?"	Helps people align the new way of working	Helps people align the new way of working
Style	Structured, methodical, and risk-averse.	Flexible, adaptive, and willing to take risks.
Outcome	Creates engagement, innovation, and long-term value.	Delivers outputs on time, within budget, and per scope.

Characteristics of IT Projects: High level of complexity, Rapidly changing technology, Intangible deliverables, High stakeholder involvement, High uncertainty and risk, Dependency on requirements accuracy.

ITSM (IT Service Management): a set of policies, processes, and practices that organisations use to design, deliver, manage, and improve IT services to meet the needs of the business and its customers. Why important? Business Alignment, Improved Service Quality, Cost Efficiency

Characteristic	Explanation	Example
Intangible	Services can't be touched or stored; they're experienced.	Cloud hosting, email access
Outcome focused	The goal is what the customer can do, not the technology used.	Streaming movies (Netflix)
Co-created value	Both provider and consumer contribute to the outcome.	University IT + Lecturer = Online teaching
Continuous Delivery	Services are ongoing, not one-off products.	IT helpdesk, cybersecurity monitoring
Risk and Cost transfer	Provider handles complexity and risk.	Using AWS instead of maintaining your own data centre

Types of IT Services: Infrastructure Services Provide the foundational hardware, software, and networking. Examples: Cloud hosting (AWS EC2), data storage, network management.

Application Services: Enable users to perform specific business tasks. Examples: Microsoft 365, Canvas LMS, Salesforce CRM. Business Services. **Integrate IT services** to deliver end-to-end business value. Examples: Payroll processing, online banking, e-

commerce operations. **Support Services**: Maintain and improve the quality of IT operations. Examples: IT Helpdesk, Service Desk, Technical Support.

Aspect	Product	Service
Nature	Tangible or defined software/hardware	Intangible and experience-based
Ownership	Customer owns or licenses	Customer uses, provider owns
Delivery	One-time or versioned	Continuous
Value Focus	Features and functions	Outcomes and experiences
Example	Buying Adobe Photoshop	Subscribing to Adobe Creative Cloud

Aspect	IT Project Management	IT Service Management
Definition	The discipline of planning, executing, and closing IT-related projects within scope, time, and budget.	The discipline of designing, delivering, and improving IT services that support business operations.
Objective	To deliver a unique product, service, or result successfully.	To ensure ongoing, reliable delivery of IT services to customers and users.
Focus	Creating or changing something (e.g., developing software, upgrading infrastructure).	Maintaining and optimizing what's already running (e.g., managing incidents, changes, and SLAs).
Primary Stakeholders	Project sponsor, project manager, development teams, business users.	Service manager, service desk, end users, customers.
Typical Roles	Project Manager, Scrum Master, Developers, QA Team.	Service Manager, Incident Manager, Change Manager, Service Desk.

IT Project Management: Focuses on temporary initiatives to deliver a specific output (e.g., new software, system upgrade).

ITSM: Focuses on ongoing delivery and support of IT services to ensure continuous value creation

Service Relationship Models: Business Oriented **Business-2-Business**: focus on efficiency, scalability, and reliability. Exa: AWS, Salesforce, Intel. **Business-2-Consumer**: To meet personal or end-user needs. B2C services emphasize user experience, personalization, and convenience. Examples: Uber, Spotify.

Business-2-Government: To enable or enhance public administration and citizen services. focuses on compliance, transparency, and long-term contracts. Exa: IBM, Thales Group, Serco, Telstra. **Customer Oriented** **Consumer-2-Consumer**: rely on secure digital infrastructure, data analytics, and trust mechanisms

Exa: eBay, Facebook Marketplace. **Consumer-2-Business**: Consumers provide data, feedback, content, or development work

Exa: YouTube (content creators earning from ads), Shutterstock (individuals selling photos), Fiverr.) **Government Oriented**

Government-2-Business: These services help companies comply with laws, access resources, or participate in public procurement

Examples: Business.gov.au, ATO Business Portal. **Government-2-Consumer**: aim to enhance public welfare, accessibility, and engagement. Exa: MyGov, Service NSW, Medicare Online,

Australian Passport Application Portal. **Government-2-Government**: either within a single level of government (e.g., local to-state) or across levels (e.g., federal-to-state). Exa: ABS-Home Affairs data sharing, GovLink, UN e-Government portal, Interpol information systems) **Other** [B2E to enhance employee productivity, engagement, and satisfaction. Exa: Microsoft Teams, Google Workspace. **B2C**: manufacturers or service providers sell

Directly To end Consumers, bypassing intermediaries like distributors or retailers. Examples: Nike, Tesla. **B2M**: both B2B and B2C, a hybrid model. **Machine-to-Machine (M2M)** : communication between devices without human intervention forms the foundation of IoT (Internet of Things). Examples: Smart meters, connected cars. **Network-to-Consumer (N2C)**. **Network-to-Business (N2B)]**

4 dimensions of Service Management

Dimension	Focus Area	Key Question
Organisation and People	Structure, culture, roles, and skills	Do we have the right people and culture to deliver value?
Information and Technology	Data, tools, and digital systems	Are our technologies enabling effective service delivery?
Partners and Suppliers	External relationships and contracts	Who helps us co-create and deliver value?
Value Streams and Processes	Activities that create value	How do we transform inputs into valuable outcomes?

Information Technology Infrastructure Library (ITIL): a globally recognised framework that provides best practices for delivering high quality IT services. It defines how IT services should be aligned with business needs, ensuring consistency, efficiency, and continual improvement.

Version	Year Released	Focus/Characteristics
ITIL v1.0	1980s-1990s	Collection of 30+ books covering IT operations.
ITIL v2.0	Early 2000s	Streamlined to focus on Service Support and Service Delivery.
ITIL v3.0	2007/2011	Introduced Service Lifecycle (5 stages) and value co-creation.
ITIL v4.0	2019	Modernised for digital transformation, Agile, DevOps, and cloud environments.

ITIL 4 Components: Service Value System (SVS) holistic view, enable value co-creation with stakeholders through IT-enabled services. **Guiding Principles** all efforts align with value creation.

Governance. Service Value Chain